

EEE-103

Additional Practice Problems

Solve the following problems by writing:

- Algorithm
- Flowchart
- C Program

Problem 1: Sum of Digits

Write an algorithm, flowchart, and C program to calculate the sum of the digits of a given number.

Example:

Input: 103

Output: $1 + 0 + 3 = 4$

Practice additional variations of this problem, such as:

- Sum only the odd digits.
- Sum only the even digits.
- Multiply only the odd digits.
- Sum/multiply digits greater than 5.
- Sum/multiply digits divisible by 2/3/5 etc.

Problem 2: Series Calculation

Write algorithm, flowchart, and C program to determine the value of the following series.

Variant 1

$$S = \frac{1^2}{1!} - \frac{3^4}{3!} + \frac{5^6}{5!} - \frac{7^8}{7!} + \cdots + \frac{n^{(n+1)}}{n!}$$

Variant 2

$$S = \frac{1 \times 2}{2!} + \frac{3 \times 4}{4!} + \frac{5 \times 6}{6!} + \cdots + \frac{(n-1) \times n}{n!}$$

Variant 3

$$S = \frac{1}{\left(\frac{1}{3!} + \frac{1}{5!} + \frac{1}{7!} + \cdots + \frac{1}{n!}\right)}$$

Hint: First calculate the denominator series, then divide 1 by the calculated value.

Problem 3: Future Value of Recurring Deposit

Write an algorithm, flowchart, and C program to calculate the future value of a recurring deposit with yearly contribution and yearly growth.

The final amount is calculated as:

$$F = C \cdot Y + \text{GrowthBonus}$$

Where

$$Y = \left(1 + \frac{r}{100}\right)^{N+1}$$

$$\text{GrowthBonus} = G \times (N-1)!$$

Where:

- C — Yearly contribution
- r — Annual growth rate (%)
- G — Growth bonus
- N — Number of years
- F — Final amount after N years

Your program should:

- a) Read C , r , G , and N from the user
- b) Display the final amount with two decimal places

Hint: First calculate $1+r/100$ then calculate the raised to the power of that value. Calculate values separately and then incorporate in the formula for final result.

Problem 4: Pattern Printing

Write C programs using nested loops to print different patterns such as:

- Square
- Rectangle
- Right Triangle
- Inverted Triangle
- Pyramid

Example:

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Problem 5: Nested Loop Based Problems

Solve different series problems using nested loops. Also try prime and factorial in a range for nested loop practice.

Problem 6: Palindrome Number

Write a C program to determine whether a given number is a palindrome or not.

Hint:

Extract each (last) digit, add it to reversed number and multiply by 10 to construct the reverse of the number.

Example:

Original number = 123

Steps:

$$Reverse = ((3 \times 10) + 2) \times 10 + 1 = 321$$

If

Original number = Reversed number

then the number is a palindrome.

Problem 7: Armstrong Number

Write a C program to determine whether a number is an Armstrong number.

An Armstrong number is a positive n -digit number that is equal to the sum of its digits each raised to the power of n .

Example:

$$153 = 1^3 + 5^3 + 3^3$$

Steps to Check Armstrong Number

1. Count the number of digits in the number (n)
2. Extract each digit
3. Raise each digit to the power n
4. Sum all the results
5. If the sum equals the original number, it is an Armstrong number

Note:

This problem is very important because it requires multiple loops:

- One loop to count digits
- One loop to extract digits and sum the raised to the power values
- Another loop to calculate powers
- It's a nested loop problem as well
- Final if else check based on calculated value